





NanoPulldown: a Local *In Silico* Pulldown Screen

AlphaPulldown-style bait/prey screening with ESMFold2-Fast on a single Apple Silicon Mac Studio

Etienne Reboul

Groupe Gibbérellines et adaptation à l'environnement
Institut de Biologie Moléculaire des Plantes (IBMP), Strasbourg

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Outline

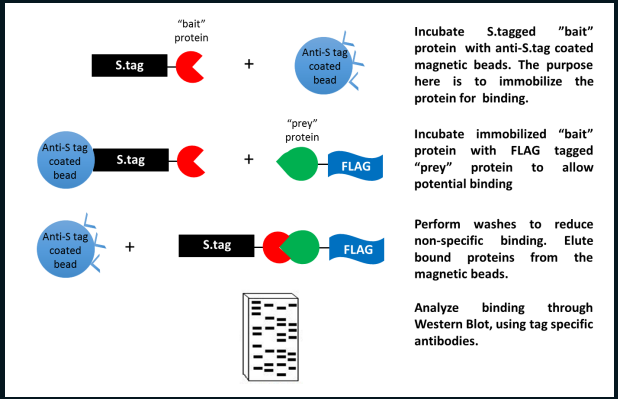
1. Introduction
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Introduction

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Pull-down assay: finding what binds a bait



Pull-down with tagged proteins (S-tag / FLAG). RaihaT, Wikimedia Commons, CC BY-SA 3.0

1. **Pull-down**: tagged *bait* immobilized on beads
2. Incubate with *prey* proteins (purified, or a lysate)
3. Wash, then elute
4. Readout: Western blot, or **mass spectrometry** (IP-MS) for a proteome-wide list of candidate preys

Limits: direct and indirect binders are mixed.

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The AlphaFold2 breakthrough

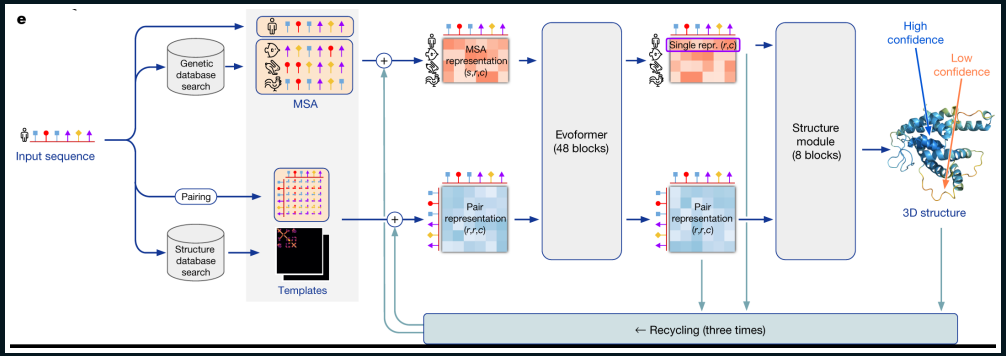


Fig. 1e, cropped; Wikimedia Commons, CC BY 4.0 (Jumper et al., 2021)

⇒ AF2 multimer is a **protein-protein docking engine** with confidence scores

John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold.
Nature, 596:583–589, 2021.
URL https://doi.org/10.1038/s41586-021-03819-2

AlphaPulldown: the workflow

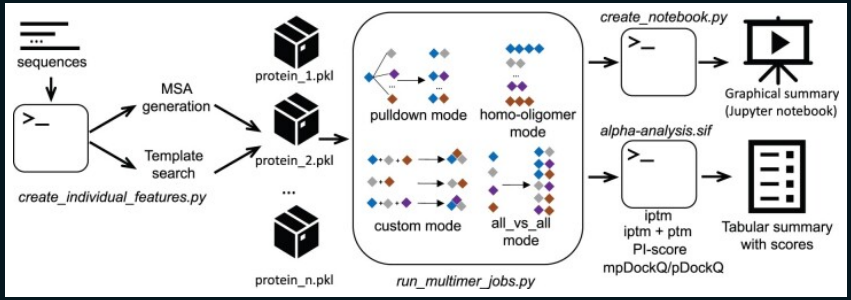


Fig. 1 of the paper; CC BY 4.0 (Yu et al., 2023)

⇒ AlphaPulldown automates identification of direct binding partners through the **in silico pulldown** with AF2-Multimer

Dingquan Yu, Grzegorz Chojnowski, Maria Rosenthal, and Jan Kosinski. AlphaPulldown – a python package for protein–protein interaction screens using AlphaFold-Multimer.
Bioinformatics, 39(1):btac749, 2023.
URL https://doi.org/10.1093/bioinformatics/btac749

Problem statement

- AlphaPulldown is **great**, but it assumes:
  - access to a GPU cluster :
    - Unmanageable to give access to a large number of users
    - Pangloss (ibmp cluster) does not have high-end GPUs for a full screen
    - High-end GPUs are expensive (around 12k EUR) currently not available through public procurement
  - MSA building :
    - Typically a slow process with Jackhmmer and HHblits
    - Requires large sequence databases (UniRef90, MGnify, BFD, PDB70, etc.)
    - Not feasible on a laptop
  - AlphaFold models as docking engine :
    - AF2 multimer is no longer State-of-the-Art (SOTA) for protein-protein docking
    - AF3 weights are available on demand but have specific licensing restrictions

$\Rightarrow$ Can we run the same screen on a single MacStudio (range price : 3k-9k EUR)?

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Protein LM as alternative folding engines

Protein LMs (e.g. ESMFold, Lin et al., 2023): **LLM-style transformers** trained on proteins.

	AF2-Multimer	**ESMFold-type (protein LM)**
Input	sequence + **MSA** + templates	**single sequence**
Evolutionary signal	read from the alignment at inference	**learned in the weights** during pre-training
Pre-processing	database searches (slow, heavy)	none
Weak spot	few homologs ⇒ poor MSA	orphan / novel sequences

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model.
Science, 379(6637):1123–1130, 2023.
URL https://doi.org/10.1126/science.ade2574

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ESMFold2-Fast

Language-model folding

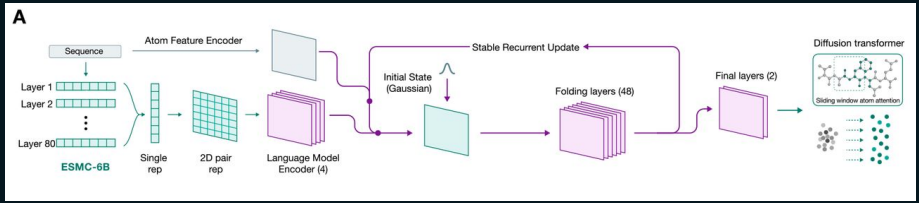


Fig. 2A, cropped, CC BY 4.0 (Candido et al., 2026)

- **ESMFold2-Fast** is designed for **high-throughput screening** and **design** of protein sequences
- **Simplified architecture**: 24 folding layers instead of 48 to reduce memory footprint and inference time
- **No MSA**: sequence-only prediction competitive with MSA-based methods (slightly worse than AF3)
- ∼9.4 s for 1 024 residues on one H100 (reported 1.3× faster than AF3)

⇒ **This is what makes a high-throughput pulldown screen on a Mac Studio feasible**

Candido, Hayes, Derry, Rao, Lin, Verkuil, et al. Language modeling materializes a world model of protein biology.
bioRxiv, 2026.
URL https://doi.org/10.64898/2026.06.03.729735

Goal

- Primary goal: Running a pulldown screen with ESMFold2-Fast on a single Apple Silicon Mac Studio
- Secondary technical goals:
  - Reproducable and scalable pipeline using Snakemake (workflow management system)
  - Easy-to-use configuration for adding new bait/prey pairs
  - Easy-to-run on a single machine (no cluster, no HPC scheduler)
  - Easy-to-read report with all the relevant metrics and figures
- Test case: BPM6 screen over the PXD013906 prey list and compare ranking against the PNAS Dataset S1 known / significant interactors (Chico et al., 2020)

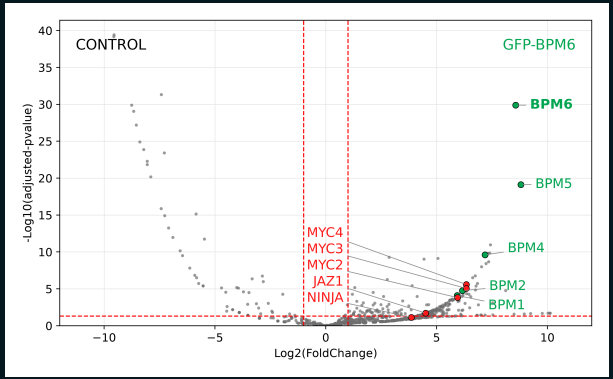
Jose Manuel Chico, Esther Lechner, et al. CUL3BPM e3 ubiquitin ligases regulate MYC2, MYC3, and MYC4 stability and JA responses. *Proceedings of the National Academy of Sciences*, 117(11):6205–6215, 2020.
URL https://doi.org/10.1073/pnas.1912199117

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Pipeline

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Test case: what the BPM6 pulldown found

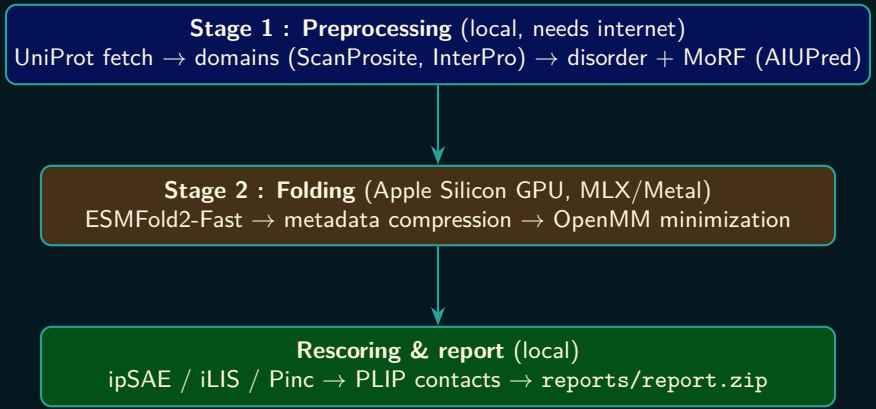


Reproduction of Fig. 1D from Dataset S1 data, CC BY (Chico et al., 2020)[a]

- **Experiment**: GFP-BPM6 immuno-precipitation from *A. thaliana* seedlings, LC-MS/MS (PXD013906)
- **Tested**: 1243 of the 1642 proteins in Dataset S1 paired with BPM6 (pairs $>$ 1000 residues excluded)
- **Green**: BPM paralogs (bait BPM6, BPM1/2/4/5) $\rightarrow$ heterodimers
- **Red**: MYC2, MYC3, MYC4, JAZ1, NINJA $\rightarrow$ candidate substrates

$\Rightarrow$ **Our benchmark**: can we find the same top interactants in the `in silico` screen?

Jose Manuel Chico, Esther Lechner, et al. CUL3BPM e3 ubiquitin ligases regulate MYC2, MYC3, and MYC4 stability and JA responses.
Proceedings of the National Academy of Sciences, 117(11):6205–6215, 2020.
URL https://doi.org/10.1073/pnas.1912199117



Everything runs on one machine, in one Snakemake call.

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Input: bait and prey lists

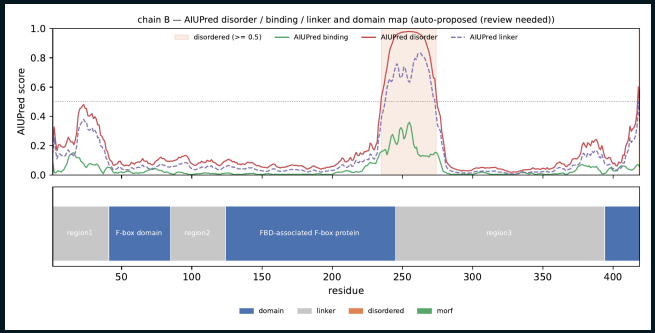

```
$ head configs/preys.txt
A0A1I9LLD2
A0A1I9LMI7
A0A1I9LNNO
A0A1I9LPP1
A0A1I9LPQ2
A0A1I9LPZ1
A0A1I9LQ33
A0A1I9LR66
...
```

- two text files of UniProt accessions: `baits.txt`, `preys.txt`
- prey list from an **IP-MS experiment** or from a **UniProt filter**
  - under investigation with Filippo BATTAGLIA: DRM2 vs. nuclear E3 ligases + F-box proteins
- a script generates all the bait × prey pairs config files (yaml)

⇒ **simple input files and one liner to add a new screen**

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Preprocessing: domains, disorder and binding regions



F4HR18 (prey of Q9M548): AIUPred disorder / binding / linker + domain map

- Automatic preprocessing of all unique sequences (baits + preys) : Domain annotation and IDR / MoRF prediction
- **IDR:** intrinsically disordered regions (long, flexible)
- **MoRF:** short disordered regions that fold upon binding

$\Rightarrow$ sequence are annotated to characterize the binding regions in interface analysis

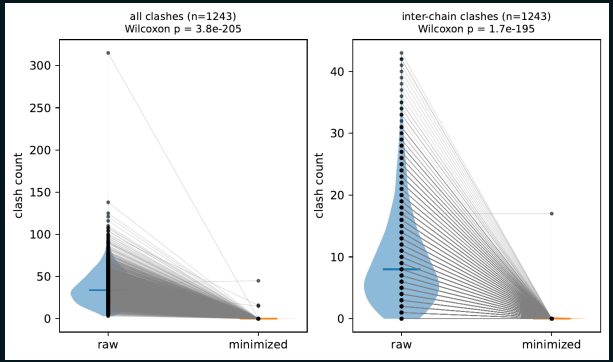
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Folding with ESMFold2-Fast on Apple Silicon

- Folding with **ESMFold2-Fast** with lower computational requirements:
  - 10 loops
  - 100 sampling steps
  - 1 seed
  - 1 diffusion sample
- **Memory usage** is the main bottleneck for folding on Apple Silicon (M1/M2/M4)
- force a **Token limit**: peak memory grows $\sim n^{2.2}$; on a 36 GB M4 Max, 1 000 tokens $\approx$ 29 GB, 145 s (`max_tokens: 1000`)
- Metadata compression to reduce disk footprint

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Minimization removes the steric clashes

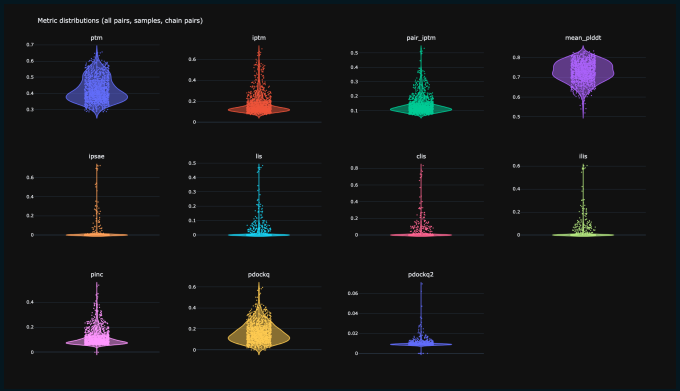


Clash count per model, raw vs. minimized (n=1243)

- Done with **OpenMM** (`amber14` + GBn2 implicit solvent; OpenCL with CPU fallback)
- Automatic Quality Control : check for divergence, NaN, or energy increase → discard the sample
- Clashes drop to ≈ 0 for nearly all models, inter-chain included (Wilcoxon $p < 10^{-190}$)

$\Rightarrow$ **Prevent false positives induced by steric clashes in the contact analysis**

Rescoring the interface

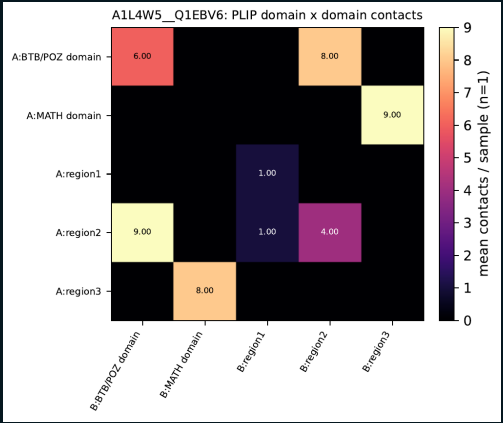


Distributions of the implemented SOTA interface metrics (ipSAE, iLIS, Pinc, pair ipTM, ...) over all pairs, samples and chain pairs

$\Rightarrow$ **Used to rank preys per bait**

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Characterizing the interface with PLIP



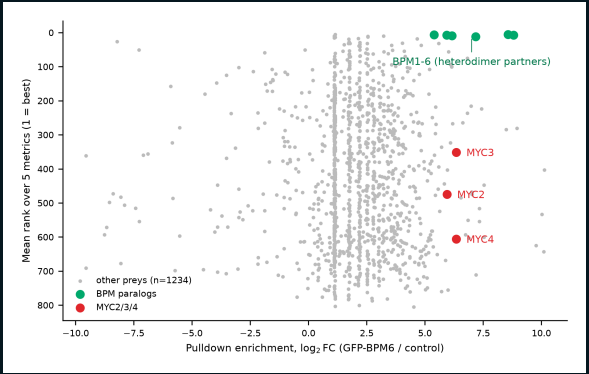
BPM6 × BPM5 (top interactor): PLIP domain × domain contacts

- **PLIP** : Protein-Ligand Interaction Profiler, detects non-covalent contacts in protein-protein interfaces
- Aggregated per **domain pair**, using the automatically annotated domains from the preprocessing stage
- **Contact types**: hydrogen bonds, hydrophobic contacts, salt bridges

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In silico pulldown results vs. BPM6 pulldown

Postprocessing



1 243 BPM6 × prey pairs; enrichment from (Chico et al., 2020)

- **Pulldown**: MYC2/3/4 strongly enriched ($\log_2$FC ≈ 6)
- **Models**: mean rank MYC3 351, MYC2 474, MYC4 606; none in the top 50 on any metric; ipSAE = iLIS = 0 for all three
- PLIP contacts mid-pack (26–37, median 28); top counts: BPM paralogs
- **Possible explanation, untested**: BPM6 binds MYC2/3 only *very weakly* in yeast two-hybrid (Chico et al., 2020); the co-IP may be bridged by other BPMs
- **Caveats**: Y2H is limited by MYC/BPM toxicity in yeast; MBP pulldown still detects BPM6–MYC. Only BPM6 run as bait

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Interactive demo

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Running it

Running the pipeline is a single Snakemake command, after creating the conda env and generating the pair configs:

```
conda env create -f envs/controller.yaml
python scripts/expand_pairs.py    # baits x preys -> configs/<pair>.yaml
# add the new pair stems to config.yaml's pairs: list
snakemake --use-conda --cores 4 --resources gpu=1 docker_heavy=3
```

⇒ **Output: one** `reports/report.zip`

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Conclusion & perspective

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Conclusion

- A **local, MSA-free pulldown pipeline**: run on a single Apple Silicon Mac Studio, no cluster required
- A **complete workflow**: Preprocessing $\rightarrow$ ESMFold2-Fast $\rightarrow$ OpenMM $\rightarrow$ rescoring $\rightarrow$ PLIP, summarized in a single zipped report
- 1 243 BPM6 $\times$ prey pairs folded and minimized end to end in $\sim$ 2.5 days on a M4 Max (36 GB) with 4 CPU cores + 1 GPU
- Adding a screen = editing the bait / prey lists + 2 or 3 commands

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Perspective

- Confirm with different bait / prey pairs, and with other known interactors
- An experimental validation of novel interactors is needed to confirm the predictions
- To expand the pipeline capabilities, we need better hardware (more GPU memory, more CPU cores) to run larger screens and longer sequences

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Thank you for your attention



Team members

Patrick ACHARD
Jean-Michel DAVIÈRE
Lali ACHARD
Olivia LEDIEU
Hans RIOS
Antoine DAVID
Étienne REBOUL

IBMP bioinformatic facility

Valérie COGNAT
David PFIEGER
Stéfanie GRAINDORGE

Collaborators

Team: Rôle de l'ubiquitine dans la régulation cellulaire

Filippo BATTAGLIA
Cyriaque CRETON
Marie-Claire CRIQUI
Pascal GENSCHIK
Esther LECHNER
Sandra NOIR
Thomas POTUSCHAK
Marlène SCHIAFFINI
Wen Hui SHEN
Vishal VARSHNEY
Agathe VERCELLETTO

Questions?

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